

INTRODUCTION

This project focuses on analyzing sales data of a coffee retail business, Afficionado Coffee Roasters, to understand customer purchasing behavior and identify important sales patterns.

In the retail industry, understanding when sales occur is as important as knowing what products are sold. Businesses often rely on assumptions such as “morning hours are busy” or “weekends generate higher sales,” but without proper data analysis, these assumptions may not always be accurate.

The main objective of this project is to perform time-based sales analysis using transaction-level data. By analyzing data across different time dimensions such as hours of the day and days of the week, we can identify peak sales periods, slow hours, and overall trends.

This analysis helps in making data-driven decisions such as optimizing staff schedules, improving store operations, and enhancing customer experience. It also allows businesses to reduce operational costs by avoiding overstaffing during low-demand periods and ensuring sufficient staff during high-demand hours.

The project uses Python, along with libraries like Pandas and Matplotlib, to clean, process, and visualize the data. Additionally, a Streamlit dashboard is developed to provide an interactive way to explore the insights.

Overall, this project demonstrates how data analysis can be used to transform raw sales data into meaningful insights that support better business decision-making.

DATASET DESCRIPTION

The dataset contains transaction-level sales data.

It includes the following important fields:

- transaction_id: Unique ID of each transaction
- transaction_time: Time of purchase
- transaction_qty: Quantity of items sold
- unit_price: Price per unit
- store_location: Store where the transaction happened
- product_category: Category of product

The dataset is clean and does not contain missing values.

ANALYSIS

In this project, data analysis was performed using Python, Pandas, and Matplotlib.

Steps performed:

- Loaded dataset using Pandas
- Converted transaction_time into datetime format
- Created a new column 'sales' by multiplying quantity and price
- Extracted time-based features like hour and day

Different types of analysis were performed:

- Yearly sales trend
- Hourly sales analysis
- Day-wise sales comparison
- Store-level performance
- Product category analysis

INSIGHTS

- Thursday is the highest sales day, indicating it is the busiest day.
- Sales peak around 10 AM, showing strong morning demand.
- Hell's Kitchen is the best-performing store among all locations.
- Certain product types contribute more to overall revenue.
- Sales vary across time, showing non-uniform customer behavior.

CONCLUSION

This project successfully analyzed sales trends and identified key patterns in customer behavior.

The insights gained from this analysis can help businesses optimize staff scheduling, improve operational efficiency, and make better decisions.

Overall, this project demonstrates the importance of data-driven decision making in retail businesses.

Executive Summary

This project analyzes sales trends of a coffee retail business using data analytics techniques.

The analysis shows that Thursday is the busiest day and sales peak around 10 AM.

The Hell's Kitchen store performs the best among all locations.

These insights can help improve staff scheduling, optimize operations, and support data-driven decision-making.